Knurling is the process of embossing a diamond shaped pattern on the surface of the workpiece. The knurling tool holder has one or two hardened steel rollers with edges of required pattern. The tool holder is pressed against the rotating work. The rollers emboss the required pattern. The tool holder is fed automatically to the required length. Knurls are available in coarse, medium and fine pitches. The patterns may be straight, inclined or diamond shaped.

The purpose of knurling is

1. To provide an effective gripping surface.
2. To provide better appearance to the work.
3. To slightly increase the diameter of the work.

1.19 Taper turning

Taper

A taper may be defined as a uniform increase or decrease in diameter of a piece of work measured along its length.



Taper turning

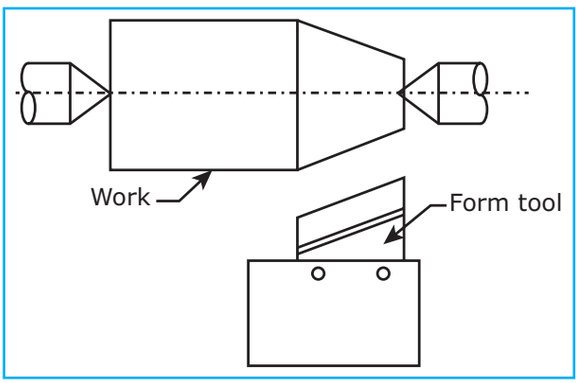
Taper turning method

1. Form tool method
2. Compound rest method
3. Tailstock set over method
4. Taper turning attachment method
5. Combined feed method

Form tool method

A broad nose tool is ground to the required length and angle. It is set on the work by providing feed to the cross-slide. When the tool is fed into the work at right angles to the lathe axis, a tapered surface is generated.

This method is limited to turn short lengths of taper only. The length of the taper is shorter than the length of the cutting edge. Less feed is given as the entire cutting edge will be in contact with the work.



Form tool method

Compound Rest Method

The compound rest of the lathe is attached to the circular base graduated in degrees, which may be swiveled and clamped at any desired angle. The angle of taper is calculated using the formula.

$$\tan\theta = \frac{D-d}{l}$$

Where

D – Larger diameter

d – Small diameter

l – Length of the taper

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